PROBLEM2:

PROGRAM STATEMENT:

Problem Statement:Progressive Electricity Billing: Calculate household bills using multiple consumption slabs, fixed charges, surcharges, and rebates, then compare bills across several months and identify unusual increases

OBJECTIVE:

To develop a Python program that calculates household electricity bills using progressive consumption slabs, fixed charges, surcharges, and rebates. The program compares bills across several months and identifies unusual increases in electricity charges.

PROGRAM:

```
# Progressive Electricity Billing Systemmonths = ["January", "February", "March", "April"]units_list = [100, 150, 180, 300]fixed_charge = 50surcharge_rate = 0.05rebate_rate = 0.10bills = []for units in units_list:    # Calculate charges using progressive slabs    if units <= 100:
energy_charge = units * 2      elif units <= 200:        energy_charge = 100 * 2 + (units - 100) * 3       else:    energy_charge = (100 * 2 +                        100 * 3 +                        (units - 200) * 5)         surcharge = energy_charge * surcharge_rate          rebate = energy_charge * rebate_rate if units < 150 else 0    total = energy_charge + fixed_charge + surcharge - rebate
bills.append(total)print("Monthly Electricity Bills")print("-------------------------")for i in range(len(months)):
print(months[i], ":", round(bills[i], 2))print("\nUnusual Bill Increases (> 30%)")for i in range(1, len(bills)):    increase = bills[i] - bills[i - 1]    percentage = (increase / bills[i - 1]) * 100  if percentage > 30:      print(months[i], "-Increase:",              round(percentage, 2), "%")
```

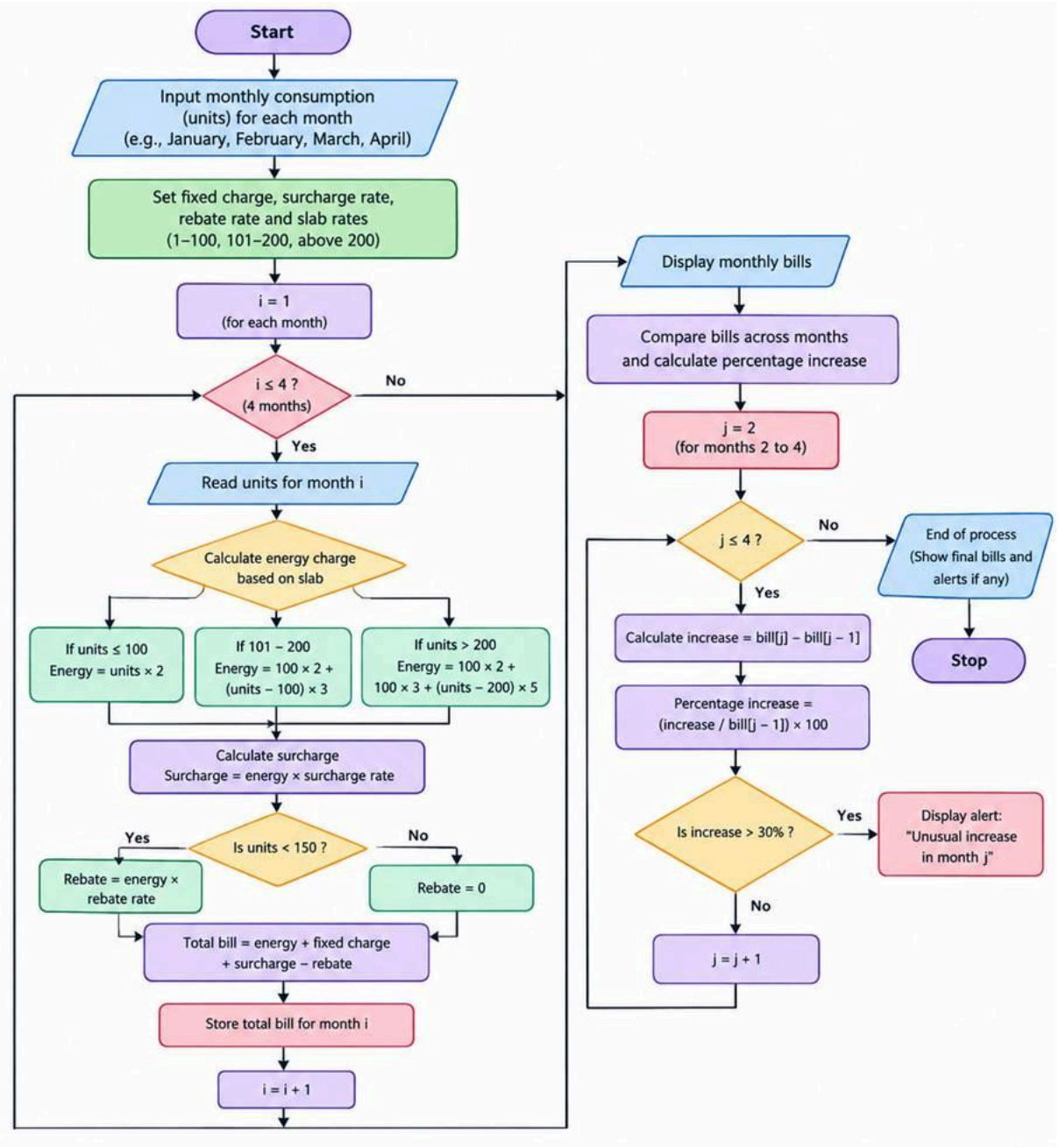
ALGORITHM:

1. Start.
2. Initialize the monthly electricity consumption values.
3. Set the fixed charge, surcharge rate, and rebate rate.
4. Read the consumption for each month.
5. Calculate energy charges using progressive slabs:
    - First 100 units: ₹2 per unit.
    - Next 100 units: ₹3 per unit.

- Above 200 units: ₹5 per unit.

6. Calculate the surcharge and applicable rebate.
7. Calculate the total bill by adding energy charges and fixed charges, then adding the surcharge and subtracting the rebate.
8. Store the total bill for each month.
9. Compare each month's bill with the previous month's bill.
10. Calculate the percentage increase.
11. If the increase exceeds 30%, display an unusual increase alert.
12. Display all monthly bills and alerts.
13. Stop.

Flowchart



PSEUDOCODE:

```
BEGIN
    months = January, February, March, April
    units = 100, 150, 180, 300

    fixed_charge = 50
    surcharge_rate = 5%
    rebate_rate = 10%

    FOR each month
    IF units <= 100
    energy = units * 2
    ELSE IF units <= 200
    energy = 100 * 2 + (units - 100) * 3
    ELSE
    energy = 100 * 2 + 100 * 3
        + (units - 200) * 5
    END IF

    surcharge = energy * 5%

    IF units < 150
    rebate = energy * 10%
    ELSE
    rebate = 0
    END IF

    bill = energy + fixed_charge
    + surcharge - rebate
    STORE bill
```

```
        END FOR

        DISPLAY all monthly bills

        FOR each month from the second month
        increase = current bill - previous bill
        percentage = increase / previous bill * 100

        IF percentage > 30
        DISPLAY unusual increase alert
        END IF
        END FOR
END
```

OUTPUT:

```
Monthly Electricity Bills
-------------------------
January : 239.5
February : 365.0
March : 459.5
April : 1025.0

Unusual Bill Increases (> 30%)
February - Increase: 52.4 %
March - Increase: 25.89 %
April - Increase: 122.96 %
```

Result

The program calculates electricity bills using progressive slabs, fixed charges, surcharges, and rebates. It compares monthly bills and flags increases greater than 30%, helping households identify unusual changes in electricity expenses.